torch.cumulative_trapezoid#

https://pytorch.org/docs/stable/generated/torch.cumulative_trapezoid.html#to

Description:

Cumulatively computes the trapezoidal rule along dim. By default the spacing between elements is assumed to be 1, but dx can be used to specify a different constant spacing, and x can be used to specify arbitrary spacing along dim. For more details, please read torch.trapezoid(). The difference between torch.trapezoid() and this function is that, torch.trapezoid() returns a value for each integration, where as this function returns a cumulative value for every spacing within the integration. This is analogous to how .sum returns a value and .cumsum returns a cumulative sum. y (Tensor) – Values to use when computing the trapezoidal rule. x (Tensor) – If specified, defines spacing between values as specified above. dx (float) – constant spacing between values. If neither x or dx are specified then this defaults to 1. Effectively multiplies the result by its value.

Function Signature

```
torch.cumulative_trapezoid(y, x=None, *, dx=None, dim=-1) →
Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

dx (float) – constant spacing between values. If neither x or dx are specified then this defaults to 1. Effectively multiplies the result by its value. dim (int) – The dimension along which to compute the trapezoidal rule. The last (inner-most) dimension by default.

Code Examples

```
>>> # Cumulatively computes the trapezoidal rule in 1D, spacing
is implicitly 1.
>>> y = torch.tensor([1, 5, 10])
>>> torch.cumulative_trapezoid(y)
tensor([3., 10.5])
```

```
>>> # Computes the same trapezoidal rule directly up to each
element to verify
>>> (1 + 5) / 2
3.0
>>> (1 + 10 + 10) / 2
10.5

>>> # Cumulatively computes the trapezoidal rule in 1D with
constant spacing of 2
>>> # NOTE: the result is the same as before, but multiplied by
2
>>> torch.cumulative_trapezoid(y, dx=2)
tensor([6., 21.])

>>> # Cumulatively computes the trapezoidal rule in 1D with
arbitrary spacing
>>> x = torch.tensor([1, 3, 6])
>>> torch.cumulative_trapezoid(y, x)
tensor([6., 28.5])

>>> # Computes the same trapezoidal rule directly up to each
element to verify
>>> ((3 - 1) * (1 + 5)) / 2
6.0
>>> ((3 - 1) * (1 + 5) + (6 - 3) * (5 + 10)) / 2
28.5

>>> # Cumulatively computes the trapezoidal rule for each row of
a 3x3 matrix
>>> y = torch.arange(9).reshape(3, 3)
tensor([[0, 1, 2],
        [3, 4, 5],
        [6, 7, 8]])
>>> torch.cumulative_trapezoid(y)
tensor([[ 0.5,  2.],
        [ 3.5,  8.],
        [ 6.5, 14.]])
```

```
>>> # Cumulatively computes the trapezoidal rule for each column
of the matrix
>>> torch.cumulative_trapezoid(y, dim=0)
tensor([[ 1.5,  2.5,  3.5],
        [ 6.0,  8.0, 10.0]])

>>> # Cumulatively computes the trapezoidal rule for each row of
a 3x3 ones matrix
>>> #   with the same arbitrary spacing
>>> y = torch.ones(3, 3)
>>> x = torch.tensor([1, 3, 6])
>>> torch.cumulative_trapezoid(y, x)
tensor([[2., 5.],
        [2., 5.],
        [2., 5.]])

>>> # Cumulatively computes the trapezoidal rule for each row of
a 3x3 ones matrix
>>> #   with different arbitrary spacing per row
>>> y = torch.ones(3, 3)
>>> x = torch.tensor([[1, 2, 3], [1, 3, 5], [1, 4, 7]])
>>> torch.cumulative_trapezoid(y, x)
tensor([[1., 2.],
        [2., 4.],
        [3., 6.]])
```

Detailed Documentation

Documentation

Cumulatively computes the trapezoidal rule along dim. By default the spacing between elements is assumed to be 1, but dx can be used to specify a different constant spacing, and x can be used to specify arbitrary spacing along dim.

For more details, please read torch.trapezoid(). The difference between torch.trapezoid() and this function is that, torch.trapezoid() returns a value for each integration, where as this function returns a cumulative value for every spacing within the integration. This is analogous to how .sum returns a value and .cumsum returns a cumulative sum.

y (Tensor) – Values to use when computing the trapezoidal rule.

x (Tensor) – If specified, defines spacing between values as specified above.

dx (float) – constant spacing between values. If neither x or dx are specified then this defaults to 1. Effectively multiplies the result by its value.

dim (int) – The dimension along which to compute the trapezoidal rule. The last (inner-most) dimension by default.